

# Full Feature Product Plan

A bigger, easier-to-read overview of the whole BaseballHelm plan: operations, stats, AutoSync, practice, scrimmage, lifting, CoachHelm, player profiles, integrations, settings, and build phases.

### Product Thesis

- BaseballHelm is the baseball-specific operating layer between existing tools and actual coaching decisions.
- The signature loop is source -> trust -> signal -> action -> player workflow -> measured outcome.

### Decision Layer

- Turns stats, practice, lifts, video, and classes into staff-owned actions.
- Built to show coaches, friends, investors, and build agents why the product is serious.

### Trusted Data Core

- Every stat, upload, and AI brief has source, scope, confidence, and raw-file lineage.
- Built to show coaches, friends, investors, and build agents why the product is serious.

### Elite Daily Workflow

- Plan practice, assign players, score scrimmages, modify lifts, and measure outcomes.
- Built to show coaches, friends, investors, and build agents why the product is serious.

| Product Pillar   | What Gets Built                                                                 | Why It Wins                                                |
|------------------|---------------------------------------------------------------------------------|------------------------------------------------------------|
| Command Center   | Today view, signals, imports, risks, tasks, video queue, staff follow-ups.      | Coaches see what matters and who owns it.                  |
| Practice OS      | Time blocks, lineups, stations, staff/player assignments, calendar publishing.  | Practice becomes measurable and connected to outcomes.     |
| Stats + AutoSync | Official XML/CSV ingest, parser profiles, confidence commits, corrections.      | Stats become trusted and useful without spreadsheet chaos. |
| CoachHelm        | Source-cited signals, reports, practice generator, meeting mode, effectiveness. | AI explains and assigns instead of just sounding smart.    |

# Market Position + Modes

BaseballHelm is not a generic team app. It is a baseball command system with mode-specific defaults.

## COOLEST HIGHLIGHT

One shared core supports college, high school, JUCO, and showcase with mode-specific navigation, onboarding, permissions, reports, and profiles.

### College Mode

- Command Center, Signals, Roster, Practice, Performance, Stats, Calendar, Meetings, Imports, Team Ops, Settings.
- Built for daily ops, official stats, player development, lifting, academics, travel, and staff accountability.

### High School Mode

- Simpler schedule, attendance, player/guardian contacts, stats, development, communication, showcase/recruiting.
- Coach-managed player accounts and optional guardian visibility without college-level admin complexity.

### Showcase Mode

- Event rosters, measurable uploads, player packets, scout/coach reports, videos, events, camps, and exports.
- Public/private profile controls plus verified/unverified labels for metrics and video.

### JUCO Mode

- College-style daily workflow with stronger transfer profile, exposure, recruiting timeline, and roster-turnover tools.
- Designed for player development plus visibility.

### Assistant Accounts

- Invite by role, position group, team, and season.
- Pitching, hitting, strength, recruiting, ops, academic, scorekeeper, video, and admin presets.

### Permission System

- Player-safe views, staff-only notes, guardian/showcase visibility, AI/source boundaries.
- Every surface knows who can see the data and who can act on it.

# Command Center + Signals

The staff cockpit: what changed, why it matters, who owns it, and what happens next.

## COOLEST HIGHLIGHT

The killer feature is the **Signal Inbox**: every source-backed insight can become a practice block, player task, lift modification, meeting topic, message, or video request.

### Coach Command Center

- Today's games, practices, lifts, classes, travel, documents, announcements, and player availability.
- Unresolved signals, import status, overdue tasks, player risk, video queue, and staff follow-ups.

### Signal Inbox

- Signals from stats, scrimmages, practice, pitch/swing data, lifts, soreness, classes, video, imports, and CoachHelm.
- Each signal has source, confidence, affected player/team, recommended owner, action, and status.

### Source Trust Drawer

- Shows official source, vendor source, coach entry, player submission, raw row/file, timestamp, reviewer, and confidence.
- No black-box AI. Coaches can see why an insight exists before trusting it.

### Action Conversion

- Turn a signal into a task, practice block, video request, lift modification, staff meeting item, message, or dev plan.
- The product tracks whether the action happened and what changed afterward.

### Staff Follow-Up Loop

- Every staff decision gets owner, due date, player, source, and outcome.
- The system avoids dead notes by making meeting items operational.

### Daily Risk Stack

- Availability, soreness, workload, class conflict, missed lift, unresolved import, video gap, and player trend flags.
- Head coach opens one screen and sees the day clearly.

# Practice + Scrimmage OS

The practice planner becomes a measurable player-development engine.

## COOLEST HIGHLIGHT

The practice builder is a signature workflow: time slots, required headline, optional details, staff/player assignments, linked signals, metrics, and calendar publishing.

### Practice Plan Builder

- Drag/reorder time blocks, resize duration, clone prior practice, use templates, or generate from CoachHelm.
- Each block has headline, description, location, staff, players/groups, equipment, signal, metric, and video.

### Calendar Attachment

- Practice plans attach to practice events and inherit date, time, location, and availability.
- Players see only their assignment cards; staff see the full plan and recap.

### Scrimmage Lineup Builder

- Drag players into P, C, 1B, 2B, 3B, SS, LF, CF, RF, DH, bench, and bullpen.
- Build batting order, substitutions, inning ranges, warnings, notes, and player availability.

### Scrimmage Modes

- Blue vs White, situational, pitcher live AB, defense-only, bullpen live, showcase testing, rain-day indoor.
- Each mode stores data as scrimmage/practice development, not official games.

### Practice Recap

- Mark blocks complete, partial, or skipped; enter station metrics; attach clips; score scrimmage; assign tasks.
- CoachHelm compares planned work, completed work, measured reps, and later outcomes.

### Practice Effectiveness

- Measures whether what was practiced improved in later practice, scrimmage, or games.
- Shows sample size, scope, confounders, confidence, and next action instead of fake certainty.

# Team Ops + Meetings

Calendar, communication, documents, classes, availability, and staff meetings become connected to baseball work.

## COOLEST HIGHLIGHT

Team management should be practical and powerful: every game, practice, lift, class, travel plan, document, message, and meeting action connects back to players and events.

### Calendar System

- Games, practices, scrimmages, lifts, meetings, travel, classes, camps, showcases, and player-only sessions.
- Attach plans, lineups, imports, docs, video, attendance, acknowledgements, and recaps.

### Communication

- Announcements, reminders, event acknowledgements, document acceptance, targeted player/staff messages.
- Messages are role-aware and event-linked, not lost in group texts.

### Class + Availability Imports

- Teamworks/class CSV or manual schedule import creates conflicts before plans go live.
- Players can be limited, unavailable, traveling, in class, injured, sore, or staff-reviewed.

### Staff Meeting Mode

- Auto agenda from signals, imports, video queue, availability, tasks, and player status.
- Every meeting item converts to owner, due date, player, and outcome tracking.

### Docs + Compliance

- Forms, PDFs, policies, travel docs, equipment checklists, acknowledgements, and staff uploads.
- BaseballHelm knows who has seen or signed what.

### Ops Command Layer

- Travel, events, availability, attendance, documents, messages, and calendar all feed the player timeline.
- Ops data becomes part of development context instead of a separate admin pile.

# Video Intelligence Layer

Video works when BaseballHelm indexes, links, hosts, and permissions it honestly.

## COOLEST HIGHLIGHT

The winning video strategy is not pretending BaseballHelm owns every vendor clip. It stores the evidence layer: clip metadata, player links, source, permissions, notes, tasks, and timeline.

### Video Index First

- Store clip source, vendor ID, URL, player, game/practice, pitch/swing, inning, count, tag, note, permissions, and evidence type.
- Even if the video lives in Synergy, 6-4-3, TrackMan, Rapsodo, OnForm, or a team drive, BaseballHelm can organize it.

### Native Uploads

- Host what BaseballHelm controls: practice clips, cage swings, bullpens, lift technique, player submissions, coach phone clips, and staff review clips.
- Generate thumbnails, tags, player timeline entries, tasks, and CoachHelm notes.

### Vendor-Linked Clips

- Synergy, 6-4-3, TrackMan media, Rapsodo video, AWRE, OnForm, and uploaded reports become linked evidence objects.
- When API/direct media is unavailable, the first build still works through URLs, exports, clip IDs, and reviewed metadata.

### Video-to-Task Workflow

- Coach opens a clip, annotates the issue, assigns it to a player, ties it to practice, and sets a follow-up.
- Player sees the clip and assignment on Player Today without staff-only context.

### Permissions + Access

- Staff-only, player-visible, position-group, team-wide, showcase-public, and private evidence modes.
- Vendor clips respect the program's access rights instead of copying video where we should not.

### Reality Check

- Definitely works: metadata, links, uploads we control, tags, tasks, timelines, practice clips, lift clips.
- Depends on subscription/API: direct vendor media pull, automated clip download, full Synergy/TrackMan/Rapsodo media sync.

# Advanced Data Flywheel

More integrations mean a richer player model, better signals, and smarter CoachHelm output.

## COOLEST HIGHLIGHT

Official stats tell what happened. Tracking tech explains movement and contact quality. Video shows evidence. Lifts and readiness add context. CoachHelm reasons across all of it.

### Rapsodo Layer

- Pitching and hitting reports feed velo, spin, movement, command, exit velo, launch, spray, session context, and attached video when available.
- If CSV/API is limited, BaseballHelm supports report/PDF/manual-reviewed imports until higher-tier export access exists.

### TrackMan Layer

- Pitch-level and batted-ball data can support pitch shape, release, location, exit velo, launch, spray, distance, and media references where enabled.
- CoachHelm can detect command decay, pitch mix shifts, contact quality trends, and fatigue patterns.

### Swing Sensor Layer

- Blast and Diamond Kinetics style data adds bat speed, attack angle, connection, on-plane efficiency, time to contact, and swing consistency.
- This separates game result problems from swing quality and timing problems.

### Video + Scouting Layer

- Synergy/6-4-3/AWRE/OnForm tags connect clips to pitch sequences, approach, scouting, short game, defense, baserunning, and player review.
- CoachHelm can cite video evidence instead of just listing numbers.

### Performance Layer

- TeamBuildr, ArmCare, bodyweight, soreness, readiness, lifts, throwing workload, and recovery status explain why a player may need modification.
- Strength and baseball workload become one decision system.

### CoachHelm Gets Smarter

- The more source-labeled data per player, the more useful the recommendations: practice blocks, video tasks, lift modifications, meeting topics, player briefs.
- It still labels confidence, sample size, scope, source, and what it cannot prove.

# Stats + AutoSync

Official files first, source-specific parsers, confidence-based commits, correction handling.

## COOLEST HIGHLIGHT

BaseballHelm becomes the trusted downstream destination for official game files. SFTP/FTP, HTTPS, postgame email, local sync agent, and manual backup feed one parser pipeline.

### AutoSync Gateway

- SFTP/FTP, HTTPS uploads, email attachments, local sync agent, official feed watcher, manual backup.

### SID Setup Wizard

- Generate credentials, invite SID/athletic communications, send simple setup instructions, run test file.

### Confidence Commit

- 98-100 auto-commit; 95-98 commit with warning; 85-95 review; under 85 hold.

### Correction Diffs

- Same hash ignored; same game with new hash becomes correction review.

| Source                  | How It Gets In                                                      | What BaseballHelm Does                                                   |
|-------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------|
| GameChanger college     | XML, season CSV, email/upload where available.                      | Official games, season stats, play-by-play if included, postgame report. |
| StatCrew/Presto/SIDEARM | XML-first parser, duplicate game-file destination, upload fallback. | Box score, fielding, catching, baserunning, corrections, season recalc.  |
| TrackMan/Rapsodo        | Pitch, batted-ball, hitting, bullpen, practice session exports.     | Pitch events, batted-ball facts, command/shape/fatigue signals.          |
| Synergy/6-4-3/AWRE      | Reports, clips, video indexes, URLs, reviewed tags.                 | Video evidence, scouting notes, timeline clips, player tasks.            |
| TeamBuilder/Teamworks   | Lift exports, classes, schedules, availability, team ops files.     | Performance history, class conflicts, readiness, player availability.    |



# Parser + Trust Engine

Every file knows what it is, where it belongs, and how much the staff can trust it.

## COOLEST HIGHLIGHT

The import screen should feel smart: provider detection, grain, target tables, player matches, warnings, preview commit, rollback, and source health.

### Parser Registry

- GameChanger, StatCrew/Presto, SIDEARM, TrackMan/Rapsodo, Synergy/6-4-3, AWRE, Blast/DK, TeamBuildr/Teamworks, CSV/PDF.

### Storage Separation

- Official games, scrimmages, practices, development sessions, lifts, readiness, classes, and videos use separate scopes.

### Import Review Screen

- Detected source, confidence, warnings, matched players, unmatched rows, conflicts, target tables.

### Source Health

- Files received, success rate, failure rate, auto-commit rate, review-needed rate, average time, correction frequency.

| Import Step | Coach Sees                                            | System Result                                                   |
|-------------|-------------------------------------------------------|-----------------------------------------------------------------|
| Detect      | Provider, file type, grain, confidence.               | Routes to correct adapter and stores raw file.                  |
| Normalize   | Mapped fields, player matches, target objects.        | Converts every source to BaseballHelm model.                    |
| Validate    | Impossible values, missing fields, totals, conflicts. | Protects trusted stats from bad uploads.                        |
| Commit      | Preview diff, changes, warnings, rollback option.     | Writes official or development tables and recalculates signals. |
| Learn       | Saved mappings and reviewed corrections.              | Future imports get faster and safer.                            |

# Elite Baseball Stats

Official, scrimmage, practice, tracking-tech, strength, readiness, video, and class context.

## COOLEST HIGHLIGHT

The Stats Center should cover everything elite programs track while never mixing official games, scrimmages, practices, and development metrics without labels.

### Official Game Stats

- Batting, pitching, fielding, catching, baserunning, box score, play-by-play, season totals, splits.

### Pitch + Swing Events

- Pitch type, velo, spin, break, location, count, swing/take, whiff, chase, zone, result.

### Batted Ball + Video

- Exit velo, launch, spray, distance, hang time, hard-hit, barrel/sweet spot if available.

### Context Stats

- Lift compliance, soreness, readiness, bodyweight, classes, attendance, task completion, video review.

| Domain           | Stats Kept                                                                                    | Best Product Use                                                    |
|------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Hitting          | PA, AB, R, H, 2B, 3B, HR, RBI, BB, K, HBP, SH, SF, GDP, SB, CS, QAB, splits.                  | Trend windows, approach signals, practice targets, player briefs.   |
| Pitching         | IP, BF, pitches, strikes, runs, ER, hits, BB, K, HBP, HR, WP, balks, inherited, saves, holds. | Workload, command decay, pitch mix, bullpen and lift modifications. |
| Defense/Catching | Errors, assists, putouts, DP, blocks, receiving, pop time, throwdowns, PB, CS, tags.          | Position-group development, catcher workload, video tasks.          |
| Baserunning      | Steals, caught stealing, reads, extra-base conversion, pickoffs, advancement, decisions.      | Situational practice blocks and CoachHelm follow-up.                |
| Development      | Scrimmage stats, practice reps, pitch/swing/batted ball, lifts, readiness, classes, video.    | Complete player snapshot and practice effectiveness.                |

# CoachHelm Intelligence

AI that explains, assigns, measures, and admits uncertainty.

## COOLEST HIGHLIGHT

CoachHelm is not a chatbot bolted onto stats. It creates source-cited baseball signals, turns them into staff actions, and measures whether the action worked.

### Signal Generation

- Find trend changes, workload risks, practice gaps, import problems, class conflicts, video evidence gaps, player risk.
- Every signal includes source refs, confidence, sample size, owner, and next action.

### Postgame AI Report

- After trusted stat commit: final score, offense, pitching, defense, workload, player follow-ups, series implications.
- Reports are built from committed or clearly labeled reviewed sources.

### Practice Generator

- Generate time-blocked plans from duration, theme, facilities, staff, player groups, signals, opponent, workload, classes, lifts.
- Coach edits before publish; players see only their assignments.

### Practice Effectiveness

- Compare focus area, players assigned, practice reps, completion quality, and later game/scrimmage/practice outcomes.
- Shows sample, confidence, confounders, and recommended next action.

### Staff Meeting Mode

- Auto agenda from signals, imports, videos, tasks, availability, player status, and staff priorities.
- Decisions become follow-ups with owners and later outcome checks.

### Honesty Rules

- No unsupported causality. No mixed scopes without labels. No pretending tiny samples are certainty.
- Allowed language: early signal, associated with improvement, not enough sample, no comparable window.

# Lifting + Performance OS

Baseball-specific strength workflows built on a future multi-sport core.

## COOLEST HIGHLIGHT

The lifting system should be first-class: preset lifts, player lift UX, bodyweight, soreness, readiness, last-used weights, progression, workload overlays, and strength dashboard.

### Strength Coach Dashboard

- Today's lift groups, completion, missed/modified lifts, soreness map, readiness alerts, bodyweight trends, exercise load progress.
- Filter by team, position group, lift group, player, risk, completion, and date.

### Preset Lift Library

- In-season maintenance, preseason strength, recovery day, lower-body power, pitcher post-outing, pre-start primer, catcher recovery, speed/agility, showcase prep.
- Templates include sets, reps, load, RPE/RIR, rest, notes, videos, and modification rules.

### Player Lift Tracking

- Players enter completion, load used, reps, RPE/RIR, missed sets, modification reason, soreness/pain, bodyweight, notes.
- The system remembers last used weights and suggests progression.

### Readiness Check-Ins

- Sleep, energy, stress, total soreness, shoulder, elbow, lower-body, back, illness, throwing readiness, lifting readiness.
- Shoulder/elbow affects throwing; lower-body affects sprint/defense/lift.

### Workload Overlay

- Pitcher, catcher, two-way, bullpen, game, throw, practice, and lift load all live together.
- CoachHelm can suggest modifications without claiming fake causality.

### Multi-Sport Core

- Generic performance sessions, templates, assignments, results, check-ins, exercises, bodyweight.
- Baseball overlays add pitcher readiness, catcher readiness, two-way workload, and practice effectiveness.

# Player Profiles + Today

Everything organized around the player, with staff depth and player simplicity.

## COOLEST HIGHLIGHT

The player profile is the deep source-aware snapshot. Player Today is the simple mobile action surface: schedule, practice assignment, lift, check-in, video, task, and development priority.

### Player Snapshot

- Official stats, scrimmage stats, practice metrics, pitch/swing/batted-ball, video, tasks, lift history, soreness, bodyweight, classes, notes.
- Every number links to scope and source.

### Player Today

- Today's schedule, practice assignments, lift, soreness check-in, tasks, videos, class/travel conflicts, announcements, acknowledgements.
- No staff-only clutter. Just clear action cards.

### Player Timeline

- Imports, games, practices, lifts, check-ins, videos, tasks, staff notes, meetings, profile bests, limitations.
- Timeline shows how development is actually happening.

### Video Evidence

- Clips tied to pitches, swings, lifts, practice blocks, scrimmages, coach notes, and player tasks.
- Coach annotations become assignments and later outcome checks.

### Classes + Availability

- Class schedules, academic conflicts, travel, illness, soreness, limitations, attendance, acknowledgements.
- CoachHelm flags conflicts before practice or lift plans are published.

### Recruiting Passport

- Verified measurables, video, event stats, scout notes, profile bests, public/private controls, packet export.
- High school, JUCO, and showcase modes get stronger exposure tools.

# One-Shot Build Plan

The feature plan becomes an execution plan: verify, build, import, AI, performance, QA.

## COOLEST HIGHLIGHT

The build agent should use this as the showable feature overview and the V2-V8 docs as the binding implementation contract. No vague dashboards, no orphaned uploads, no unsourced AI.

### Settings

- Program type, teams, seasons, roster status, logo, feature flags, player policy, public profile, demo mode.

### Data Architecture

- Programs, teams, players, events, games, practices, stat sources, import runs, raw files, signals, tasks, performance, check-ins, video.

### Demo Seed System

- College, high school, showcase, and JUCO sample programs with rosters, games, stats, plans, lifts, classes, video, signals.

### QA Gates

- RLS, parser tests, duplicate/correction tests, source drawer checks, mobile player UX, role permissions, Playwright smoke, visual QA.

| Phase                       | Primary Outcome                                                                                               | Must Include                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Phase 0 - Verify            | Confirm current repo routes, Supabase types, migrations, auth/RLS, baseball tables, upload flow, UI patterns. | No blind rewrites. Reconcile existing app structure first.                          |
| Phase 1 - Core OS           | Command Center, roster, player profile, calendar, practice builder, roles, settings, demo data.               | Professional UI, player mobile Today, assistant accounts, program modes.            |
| Phase 2 - Stats/Imports     | Parser registry, raw file storage, official/scrimmage/practice separation, review queue, source trust.        | GameChanger/StatCrew/Presto paths, generic CSV/XLSX, duplicate/correction handling. |
| Phase 3 - CoachHelm         | Signals, postgame reports, practice generation, meeting mode, action conversion, practice effectiveness.      | Source citations, confidence, honesty rules, outcome tracking.                      |
| Phase 4 - Performance/Scale | Lifting OS, readiness, video evidence, recruiting passport, AutoSync expansion, monitoring.                   | Strength dashboard, player lift UX, class conflicts, multi-sport core.              |